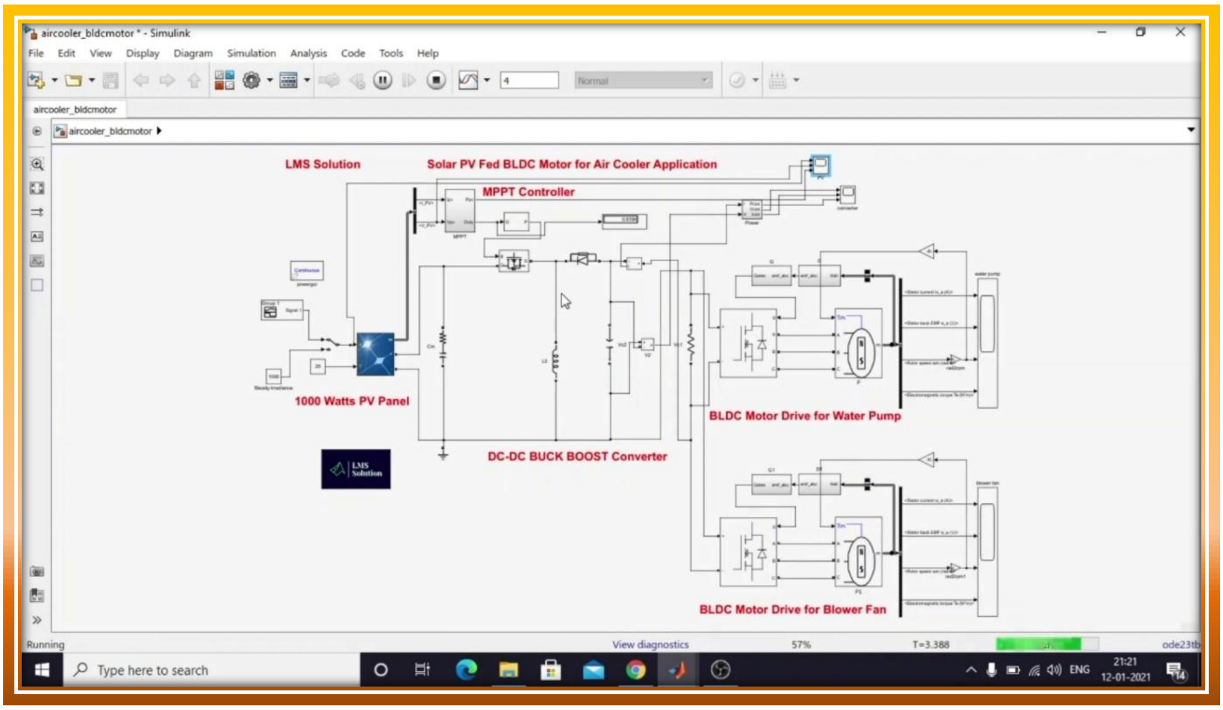
Technical Whitepaper

Design and Performance Analysis of a Solar Photovoltaic-Fed BLDC Motor Drive for Cooling Applications



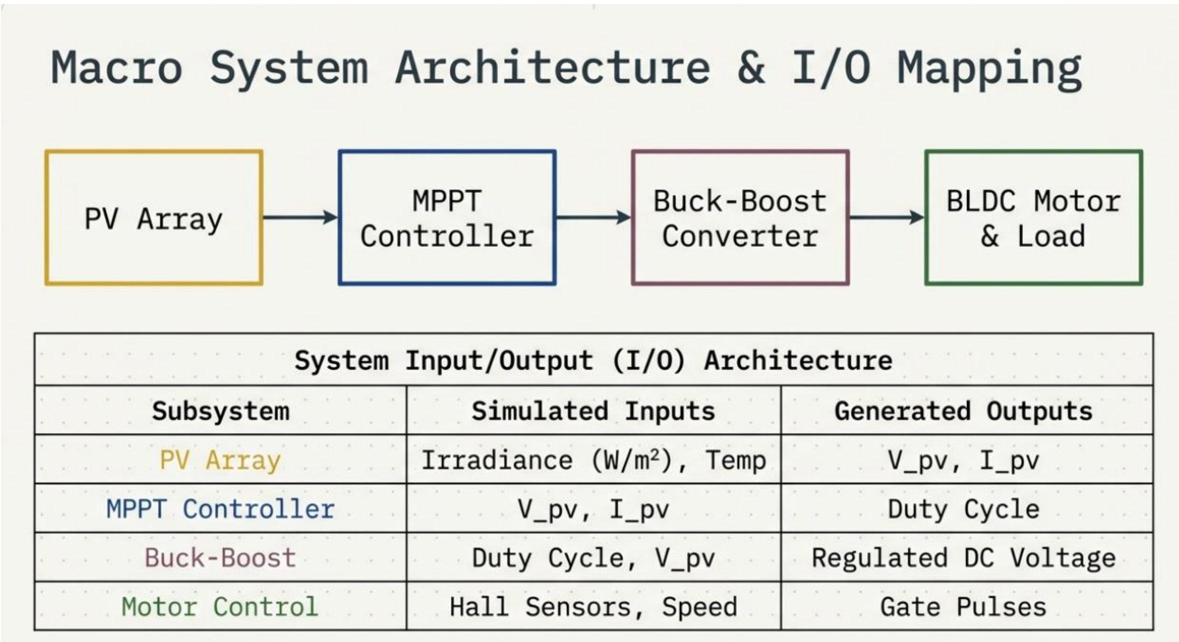
Abstract

This study presents a comprehensive design and performance analysis of a 1000 W solar photovoltaic (PV)-fed Brushless DC (BLDC) motor drive optimized for cooling applications, such as water pumps and blower fans. A critical challenge in standalone solar systems is the non-linear output of PV arrays under fluctuating atmospheric conditions. To address this, a DC-DC Buck-Boost converter is employed, regulated by a Perturb and Observe (P&O) Maximum Power Point Tracking (MPPT) algorithm. This configuration ensures peak power extraction while providing the voltage flexibility required to maintain a stable DC bus. The BLDC motor is driven via a three-phase Voltage Source Inverter (VSI) using electronic commutation derived from Hall-effect sensor feedback. Simulation results obtained

through MATLAB/Simulink validate the system's robustness under dynamic irradiation transients ranging from 1000 W/m² to 400 W/m². The analysis confirms high tracking accuracy and stable motor performance across variable mechanical loads. The integration of robust MPPT control with precise electronic commutation demonstrates a reliable and efficient architecture for sustainable, off-grid cooling infrastructure.

Keywords

Brushless DC (BLDC) Motor; Buck-Boost Converter; Maximum Power Point Tracking (MPPT); MATLAB/Simulink; Perturb and Observe (P&O); Solar Photovoltaic (PV) System.

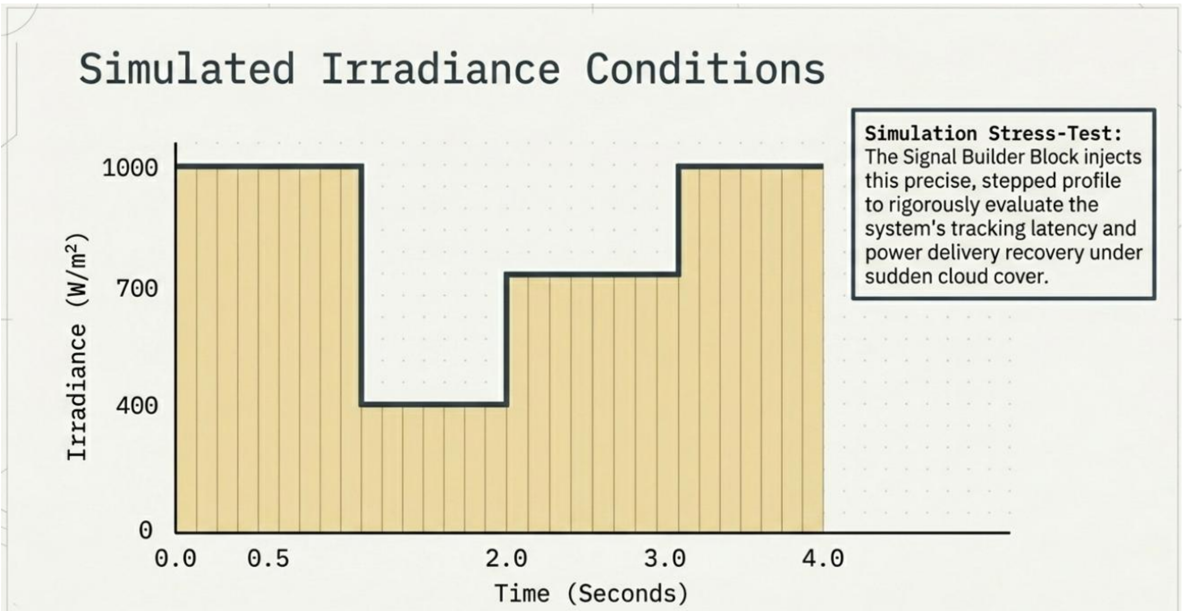


System Input/Output (I/O) Architecture		
Subsystem	Simulated Inputs	Generated Outputs
PV Array	Irradiance (W/m²), Temp	V_pv, I_pv
MPPT Controller	V_pv, I_pv	Duty Cycle
Buck-Boost	Duty Cycle, V_pv	Regulated DC Voltage
Motor Control	Hall Sensors, Speed	Gate Pulses

I. Introduction

The strategic integration of renewable energy into small-scale residential and industrial sectors has become a pivotal research area in power electronics. Solar photovoltaic (PV) systems, in particular, offer a sustainable energy source for driving air coolers and water pumps in remote locations. As global emphasis shifts toward energy efficiency, the transition from conventional AC induction motors to Brushless DC (BLDC) motors has accelerated. This shift is driven by the BLDC motor's high power density, superior torque-speed characteristics, and reduced maintenance requirements due to the absence of mechanical brushes.

The primary objective of this research is to evaluate the design and performance of a solar-fed BLDC drive system. While solar energy provides a clean power source, its inherent intermittent nature presents significant control challenges, particularly during rapid cloud cover or atmospheric shifts. The research gap addressed herein lies in the necessity for a control framework that maintains operational stability and maximum power extraction during these transients. By synthesizing a robust Maximum Power Point Tracking (MPPT) strategy with efficient electronic commutation, this study investigates the dynamic response of the system under variable irradiation. The following sections detail the mathematical modeling, control logic, and simulation-based validation of the proposed drive.

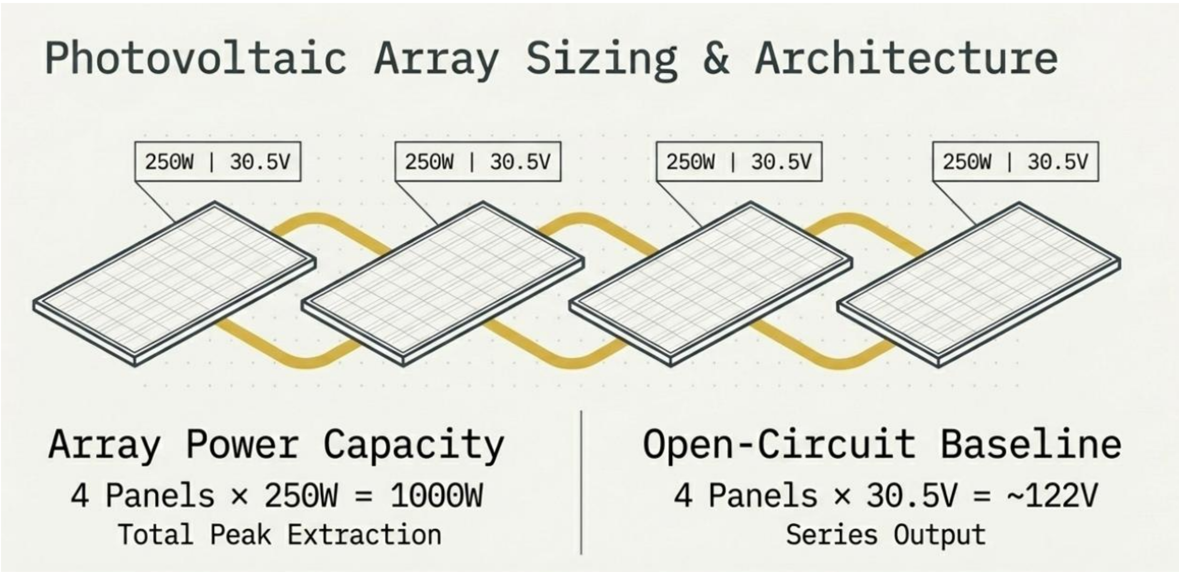


II. System Configuration and Mathematical Modeling

The proposed system architecture comprises a PV array, a DC-DC Buck-Boost converter, a three-phase VSI, and a BLDC motor. This modular configuration allows the PV source to operate at its maximum power point while the Buck-Boost converter regulates the voltage to meet the motor's requirements, regardless of whether the PV output voltage is higher or lower than the nominal DC bus voltage.

PV Array Configuration

The system utilizes a 1000 W PV array consisting of four 250 W monocrystalline panels connected in series. Each panel features a maximum power point voltage $(V_{mp})$of 30.5 V, resulting in a total string voltage of approximately 122 V at standard test conditions.

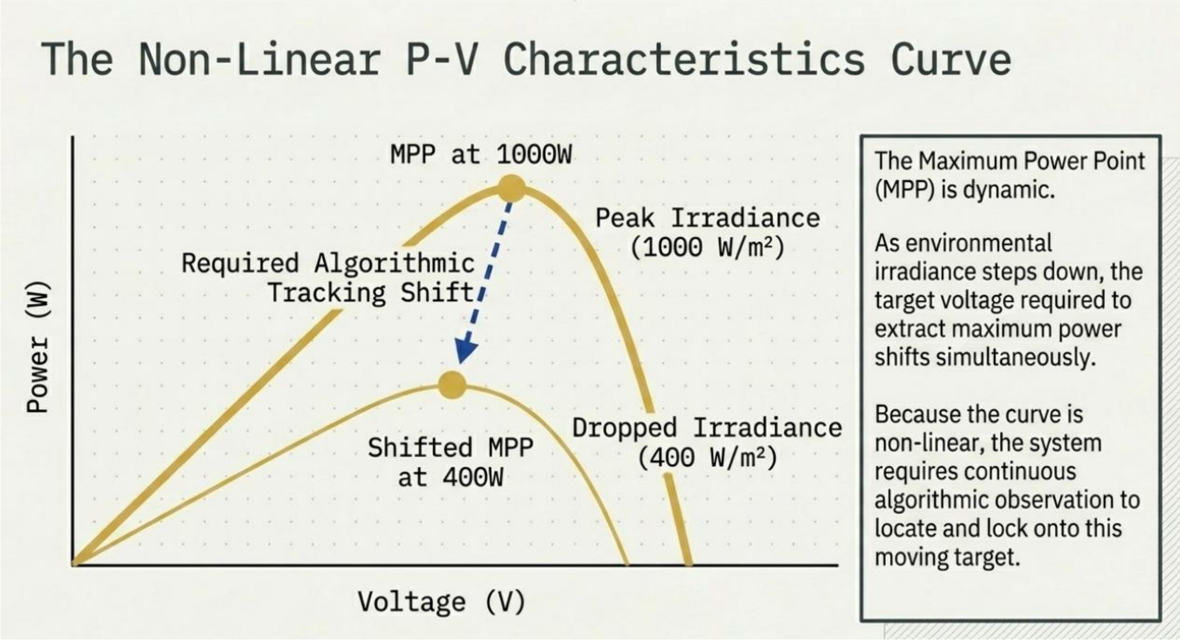


To meet the mechanical load demands of the air cooler, the array is scaled using four 250-watt panels in series, establishing a 122V nominal baseline prior to power conversion.

Mathematical Representation of the PV Cell

To accurately simulate the system, the non-linear behavior of the PV cell is modeled using the standard single-diode equation. The output current $(I)$is defined as:

$$I = I_{ph} - I_o \left[\exp\left(\frac{q(V + IR_s)}{nkT}\right) - 1\right] - \frac{V + IR_s}{R_{sh}}$$



where,

I is the PV output current,

I_{ph} is the photocurrent,

I_o is the reverse saturation current,

V is the PV output voltage,

R_s is the series resistance,

R_{sh} is the shunt resistance,

q is the electron charge,

k is Boltzmann's constant,

T is the cell temperature, and

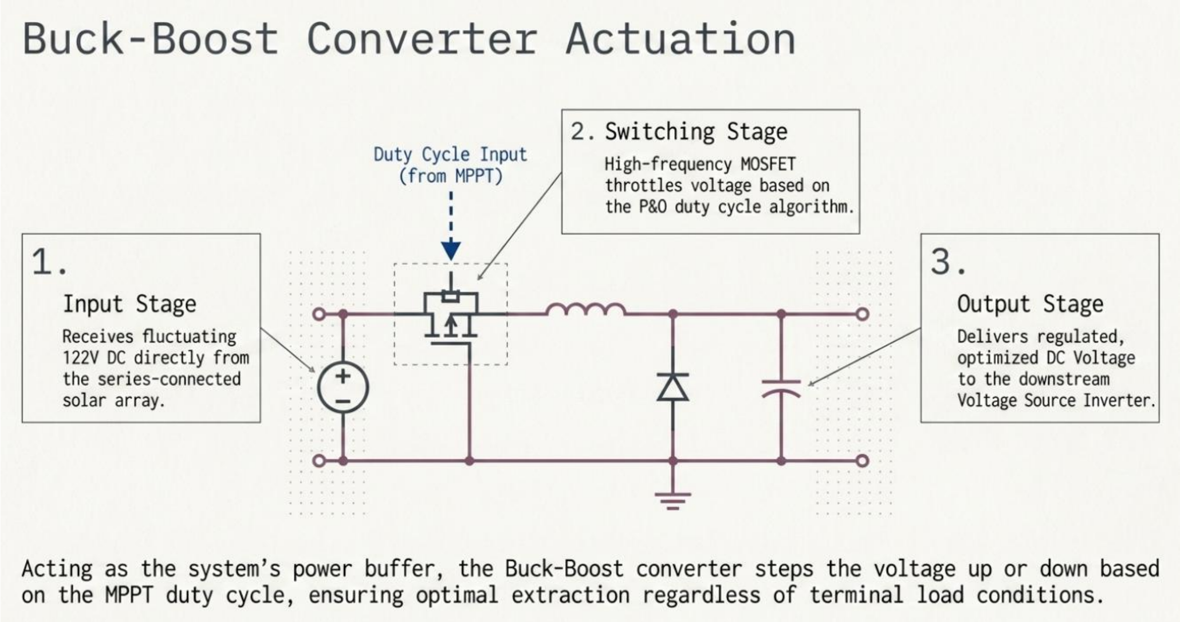
n is the ideality factor.

This equation represents the non-linear current-voltage characteristics of the PV cell by considering the photocurrent generation, diode current, series resistance loss, and shunt leakage current.

Buck-Boost Converter Dynamics

The Buck-Boost converter acts as the impedance-matching interface between the PV source and the inverter. Its role is to adjust the effective load seen by the PV array by varying the duty cycle (D). The relationship between the input voltage $(V_{in})$and the output voltage $(V_{out})$is given by:

$$V_{out} = V_{in}\left(\frac{D}{1-D}\right)$$



where,

V_{out} is the output voltage of the Buck-Boost converter,

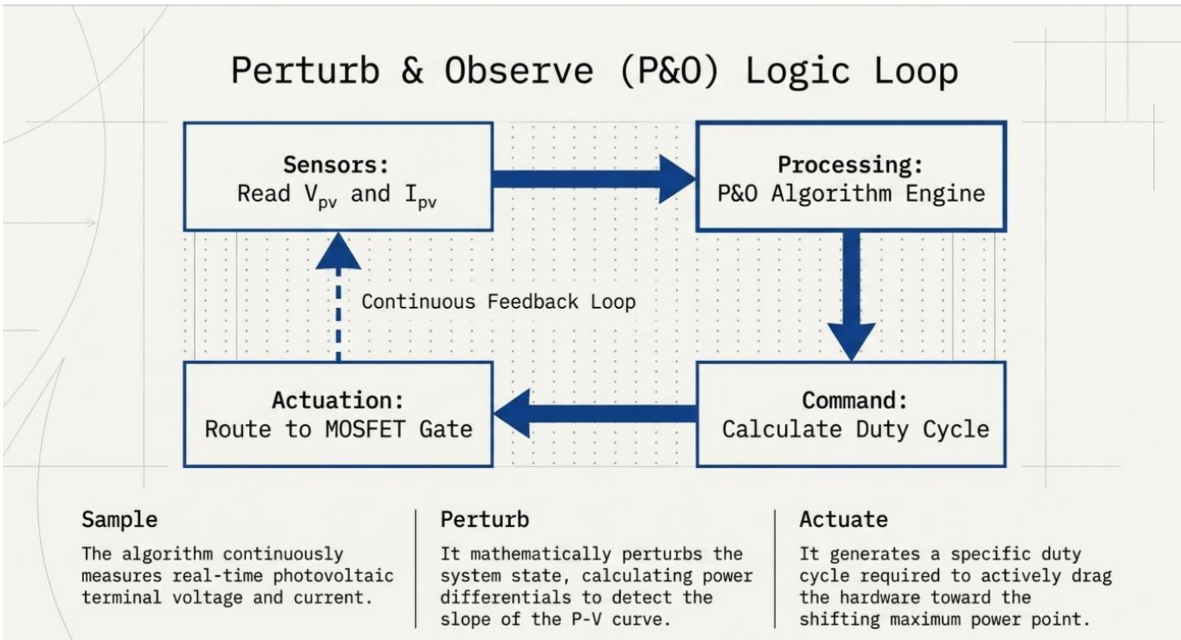
V_{in} is the input voltage from the PV array, and

D is the duty cycle of the converter.

By modulating Dvia the MPPT controller, the system maintains the PV voltage at V_{mp}, ensuring that the maximum possible power is transferred to the DC link for the BLDC drive.

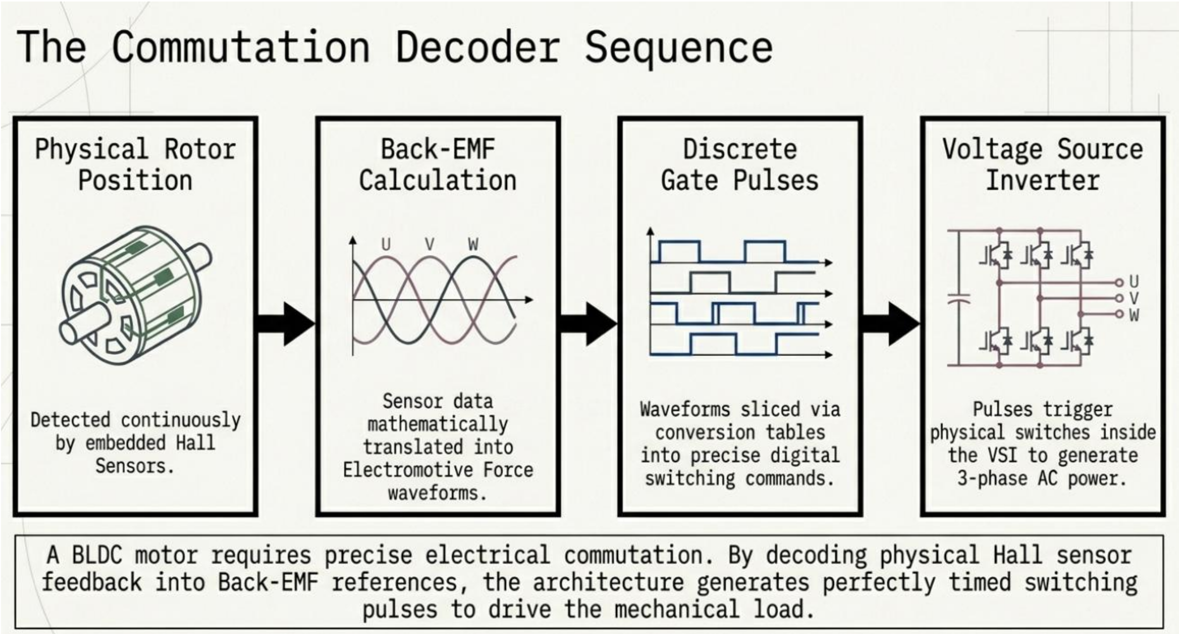
III. Proposed Control Strategy

The system utilizes a dual-control topology: the source-side control optimizes energy harvesting via MPPT, while the load-side control manages the BLDC motor's commutation.



P&O MPPT Algorithm: Technical Evaluation

The Perturb and Observe (P&O) algorithm is selected for its simplicity and effectiveness in tracking the Maximum Power Point (MPP). However, the technical significance of its implementation lies in the critical trade-off between perturbation step size and system stability. A large step size enables rapid tracking during the 1000 W/m² to 400 W/m² transitions, but results in increased steady-state oscillation around the MPP. Conversely, a small step size reduces oscillations but slows the transient response. This study utilizes an optimized step size that balances tracking speed with minimal power ripple, ensuring the 1000 W array operates at peak efficiency despite the non-linearities of the P-V curve.



BLDC Motor Commutation Logic

The BLDC motor requires electronic commutation to synchronize the stator currents with the rotor position. This is implemented using 120-degree conduction mode logic based on signals from three Hall-effect sensors. The Hall signals are converted into Back-Electromotive Force (Back-EMF) references, which then dictate the switching sequence for the six MOSFETs in the VSI.

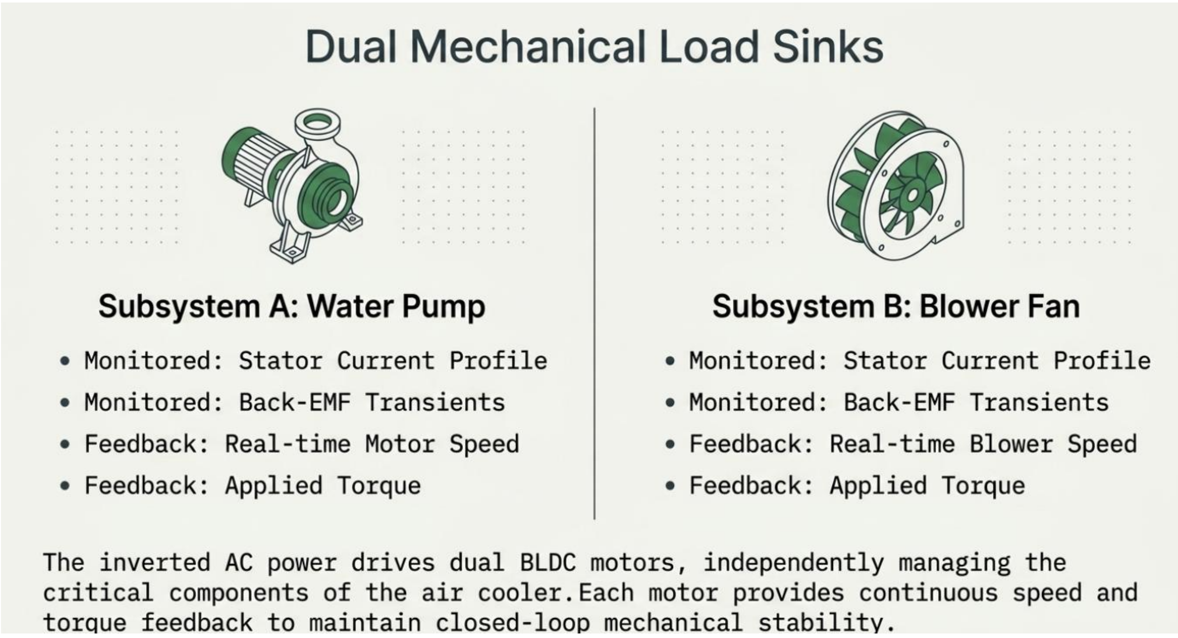
The following table summarizes the 120-degree conduction logic utilized to generate gate pulses.

Table 1. 120-Degree Conduction Logic for BLDC Motor Drive

Hall Sensor A	Hall Sensor B	Hall Sensor C	Back-EMF Phase A	Back-EMF Phase B	Back-EMF Phase C
1	0	1	1	-1	0

0	0	1	1	0	-1
0	1	1	0	1	-1
0	1	0	-1	1	0
1	1	0	-1	0	1
1	0	0	0	-1	1

This logic ensures precise magnetic field orientation, which is essential for maintaining torque consistency in applications such as water pumps and blower fans.



IV. MATLAB/Simulink Implementation and Parameters

MATLAB/Simulink provides a high-fidelity environment for simulating the non-linear dynamics of power electronic converters and machine drives. The model allows for the simultaneous evaluation of MPPT efficiency and motor torque-speed performance.

System Parameters

The simulation parameters are based on the specifications detailed in the table below.

Table 2. Simulation Parameters of the Proposed PV-Fed BLDC Motor Drive

Parameter	Specification
PV module power	250 W
Total PV capacity	1000 W
PV array configuration	4 modules connected in series
Maximum power voltage per panel (V_{mp})	30.5 V
Total PV string voltage	122 V
Converter type	DC-DC Buck-Boost converter
MPPT method	Perturb and Observe (P&O)
Inverter type	Three-phase Voltage Source Inverter
Motor type	Brushless DC motor
Commutation method	Hall sensor-based 120-degree conduction
Target applications	Water pump / Blower fan

Variable Irradiation Profile

The system was subjected to a variable irradiation sequence to assess transient resilience.

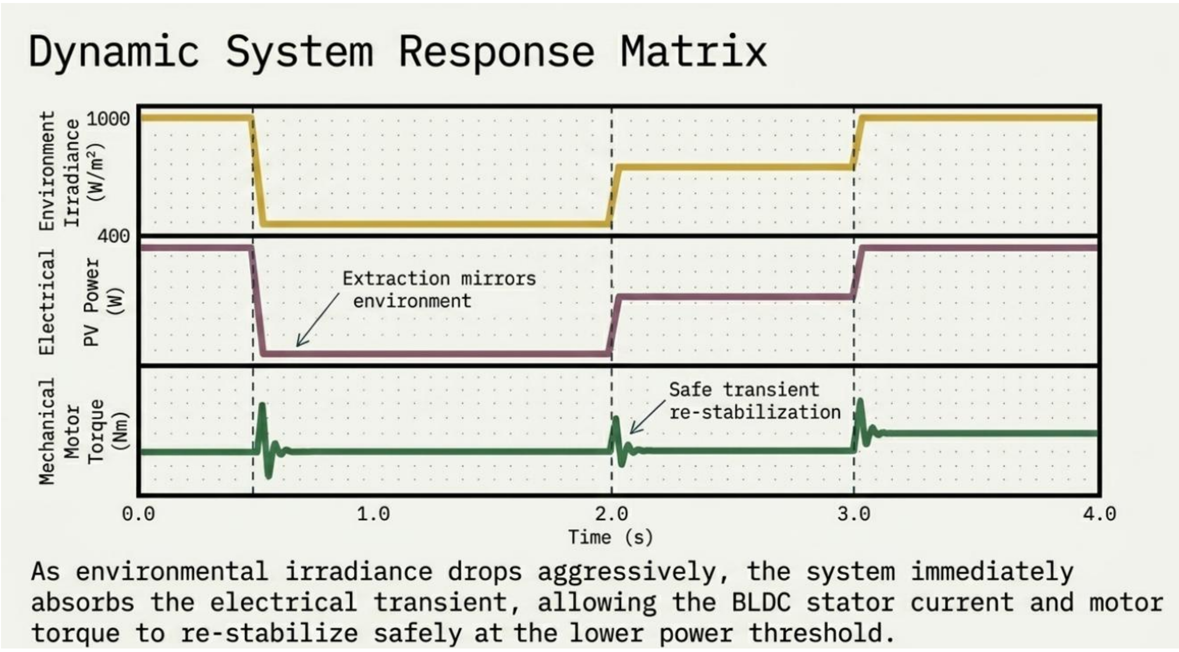
Table 3. Variable Irradiation Profile Applied in Simulation

Time Duration	Irradiation Level	Operating Condition
0.0 s – 0.5 s	1000 W/m²	Standard peak irradiation
0.5 s – 2.0 s	400 W/m²	Low irradiation / cloud simulation
2.0 s – 3.0 s	700 W/m²	Partial recovery condition
3.0 s – 4.0 s	1000 W/m²	Full recovery condition

This irradiation profile is used to verify the performance of the MPPT controller and the BLDC motor drive under sudden solar power variations.

V. Results and Discussion

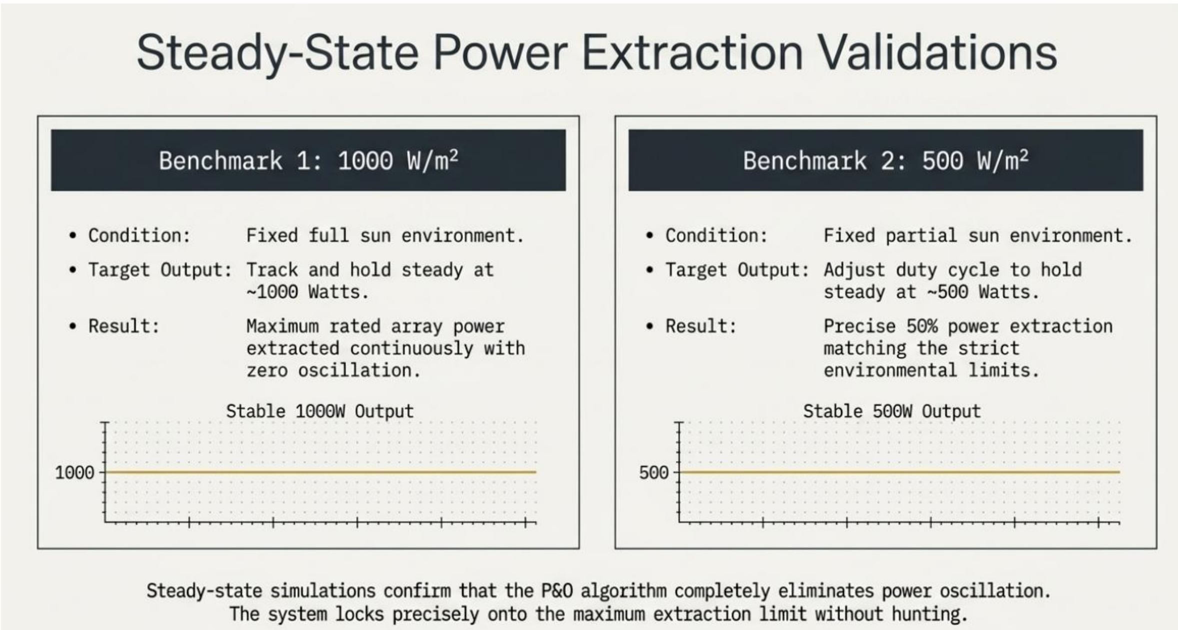
The analysis of dynamic responses during atmospheric transients is vital for verifying system stability. The simulation captures the transition of power, voltage, and motor metrics across the tested irradiation profile.



PV Power and MPPT Transient Analysis

The results demonstrate that the P&O algorithm successfully tracks the MPP throughout the irradiation steps. When irradiation drops to 400 W/m², the PV power accurately settles at the corresponding reduced power level. A significant observation is the settling time, where a finite lag occurs after each step change as the P&O algorithm searches for the new maximum power point. This transient delay is a function of the perturbation frequency, but the system demonstrates high stability once the new peak is reached.

The MPPT control maintains the PV operating point close to the peak of the P-V curve. During the transition from 1000 W/m² to 400 W/m², the PV output power decreases according to the available solar input. When the irradiation increases again to 700 W/m² and then returns to 1000 W/m², the controller adjusts the duty cycle of the Buck-Boost converter to restore maximum power extraction. This confirms that the MPPT algorithm provides reliable tracking under both decreasing and increasing irradiation conditions.



Motor Performance and Load Differentiation

The BLDC motor's response is highly dependent on the power delivered by the Buck-Boost stage. Since the motor is directly supplied from the solar-fed converter and inverter system, any reduction in PV power affects the DC-link voltage, stator current, electromagnetic torque, and rotor speed.

Load Impacts

The simulation accounts for two distinct load profiles. The water pump exhibits centrifugal load characteristics where torque is proportional to the square of the speed. This relationship can be expressed as:

$$T_L \propto \omega^2$$

where,

T_L is the load torque, and

ω is the rotor angular speed.

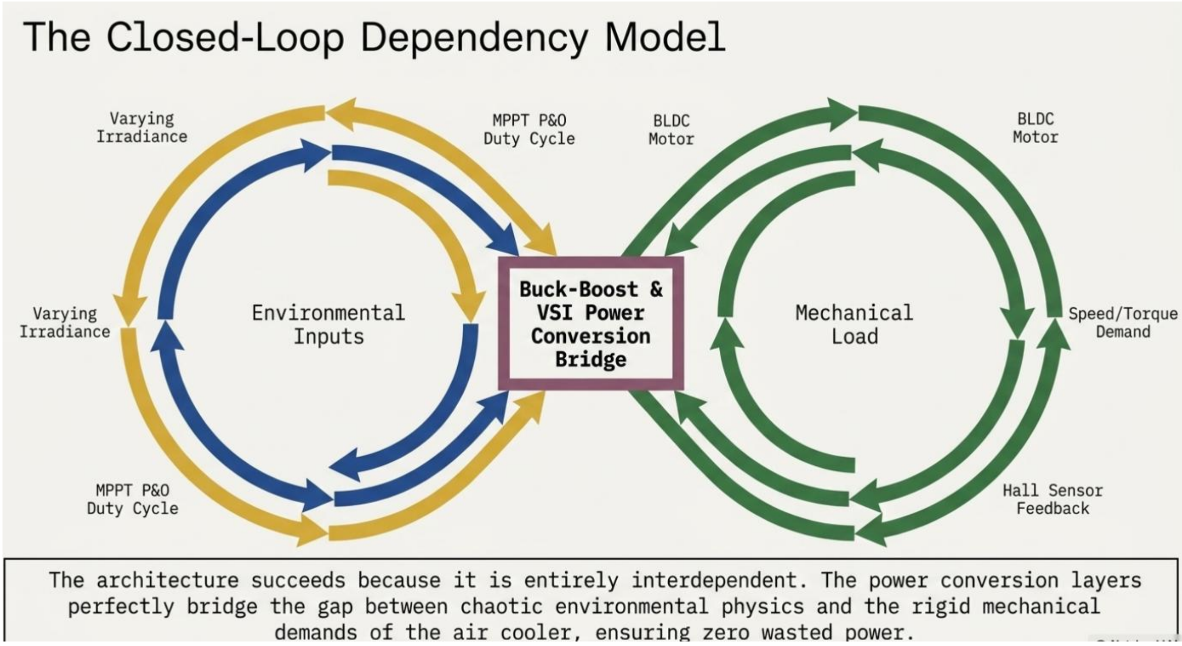
In contrast, the blower fan presents a different damping profile. During the dip to 400 W/m², the speed of both loads decreases significantly. However, the water pump exhibits a more pronounced reduction in torque due to its quadratic relationship with speed.

Current and Back-EMF Response

The stator current amplitude scales with irradiation and available PV power. When irradiation decreases, the current supplied to the inverter and motor is reduced. During high irradiation, the stator current increases to support higher torque and speed operation. The Back-EMF maintains a trapezoidal profile, validating the 120-degree commutation logic used in the VSI. This confirms that the Hall-effect sensor-based switching sequence maintains proper synchronization between rotor position and stator current excitation.

VI. Conclusion and Future Scope

This study has successfully validated a solar PV-fed BLDC motor drive architecture using a P&O-controlled Buck-Boost converter. The system demonstrates high reliability for off-grid cooling applications.



Critical Findings

1. **MPPT Tracking Accuracy:**

   The P&O controller effectively maintained maximum power extraction across variable irradiation levels from 1000 W/m² to 400 W/m², showing a robust, although finite, settling time during transitions.

2. **BLDC Motor Responsiveness:**

   The Hall-effect commutation logic ensured that the motor remained synchronized even during significant power drops, maintaining operational integrity for both centrifugal water pump and blower fan loads.

3. **System Integration Efficiency:**

   The Buck-Boost stage successfully bridged the gap between fluctuating PV output and motor requirements, providing the necessary voltage flexibility for a stable drive system.

Future Research

Further enhancements could involve the integration of battery energy storage to mitigate the speed drops observed during low irradiation periods. Additionally, implementing advanced control algorithms such as Fuzzy Logic or Artificial Neural Networks for MPPT could further reduce settling time and steady-state oscillations, leading to improved mechanical longevity of the cooling system.

The proposed MATLAB/Simulink model serves as a rigorous framework for the deployment of high-efficiency, solar-powered motor drives in sustainable infrastructure.

VII. YouTube Video

https://www.youtube.com/watch?v=shTebeWrGZ8

VIII. Purchase link of the Model

SKU: 0032
https://www.lmssolution.net.in/product-page/solar-pv-fed-buck-boost-converter-based-bldc-motor-for-air-cooler-application